

Machine Learning-based Electric Vehicle User Behavior Prediction

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Abstract—Electric Vehicles (EVs) have grown rapidly over the past decade but the integration of large-scale electric vehicles into the power grid is still a challenge. Overloading and harmonic distortions because of EV charging may impact the grid stability and may create grid issues. Power industry needs to get ready for the EV roll-out. One of the key actions to prepare for greater EV penetration and mitigate grid issues is smart charge scheduling of EVs to manage the charging load efficiently. Charge scheduling of EVs highly depends on EV users' behavior, so prediction of users' behavior is critical. Two of the important parameters, i.e., energy consumption and session duration are predicted with the Mean Absolute Error (MAE) of 1.45 kWh and 66.5 minutes respectively. These predictions are based on weekly, monthly and yearly historical public charging dataset of Adaptive Charging Network (ACN) implemented at Caltech USA. Ensemble machine learning models with base models as Random Forest (RF), Support Vector Machine (SVM) and XGBoost are used to improve the results. In contrast with the existing literature, this work utilizes the predicted session duration and weather data from local weather station along with the nearby events data and public holidays. Prediction errors are reduced in comparison to existing works with the improved data processing pipeline and feature selection.

Index Terms—Electric Vehicles (EVs), Machine Learning, Smart Charge Scheduling, Smart Grid, Charging behavior.

I. INTRODUCTION

Electric Vehicles (EVs) are getting attention not only because of their ability to reduce greenhouse gas emissions and make the environment pollution free or due to the high price of fossil fuels but also because they are 4 times more energy-efficient and have 50 times fewer moving parts compared to Internal Combustion Engine (ICE) Vehicles [1]. This makes them more economical in the long run. It is expected that the number of electric vehicles on the road will reach 140 million by 2030 which will incur an additional power demand of 500TWh from the electric grid [2].

The growth of EVs will require additional charging infrastructure and The expansion of the electric grid is not a solution. It can be handled in better ways. One way is to introduce charge scheduling of electric vehicles to coordinate between EVs and the grid. It improves the voltage profile as shown in Fig. 1 and reduces the transmission losses caused by the charging of EVs [3].

The voltage profile presented in Fig. 1 is obtained by running power flow at each 5th min interval using the pandapower

python library corresponding to the load data of 7th May 2019 acquired from the charging station located at Caltech USA. To represent network configuration of the charging station, the power grid is configured as standard Pandapower small branches-out test network [4]. The voltage regulation of the farthest bus is 2% below the nominal voltage for the base load. By adding uncontrolled EV charging (UNC) load, it drops drastically below the nominal voltage, by 16%. With the Least Laxity First (LLF) charge-scheduling algorithm from ACN-sim [5], the EVs are scheduled based on their laxity value at every 5th min, which gives only about 12% voltage drop from nominal voltage value.

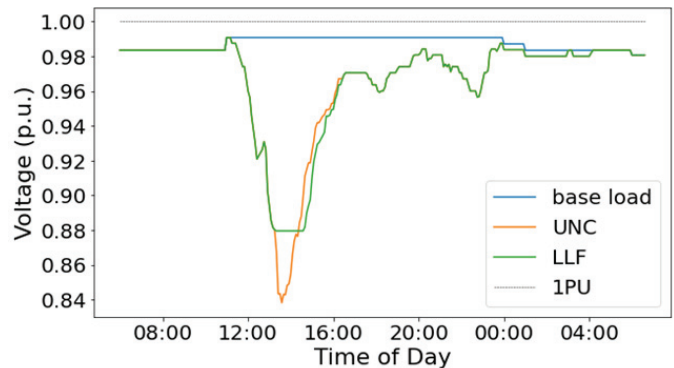


Fig. 1. Grid voltage variation for controlled and uncontrolled charging

The biggest challenge in charge scheduling of EVs is the uncertain behavior of EV users. That can be addressed using model-free nature of scheduling algorithms using reinforcement learning which doesn't require any transition probability distribution of EVs [6]. Another way is to predict the user behavior well in advance by supervised machine learning algorithms and use it for scheduling [7]–[12].

Considering the recent success of machine learning and deep learning in other fields, it can also be applied to EV user behavior prediction. Even if it predicts user behavior with some error, it can still save on charging costs for the grid as well as EV users as reported in [9]. In most of the literature, EV user behavior is determined by the following two parameters:

- 1) **Session duration:** Time duration for which the electric vehicle is plugged-in to the charging station.
- 2) **Energy consumption:** Energy consumed by the vehicle during the session duration.

In [10], ensemble machine learning employing Random Forest (RF) and Diffusion-based Kernel Density Estimator (DKDE) is used to predict the session duration and energy consumption. The residential and public charging datasets are used together and Symmetric Mean Absolute Error (SMAPE) of 10.4% for the duration and 7.5% for the energy consumption are reported. In [11], regression model, XGBoost, linear regression and Artificial Neural Network (ANN) is used to predict session duration using historical charging dataset. It has claimed Mean Absolute Error (MAE) of 82 minutes (mins). In [13], XGBoost model performs better out of many regression models used and reports an R^2 score of 0.52 and MAE of 4.6 kWh. In [14], the authors have used university campus data to predict energy consumption prediction for the next day using K-Nearest Neighbor (k-NN) and claimed SMAPE error of 15.3% for k value of 1. In [15], a novel non-residential EV charging dataset is used to predict session duration and energy consumption given arrival time distribution employing Gaussian Mixture Models (GMM) and reported SMAPE 14.4% for session duration and 15.9% for energy consumption. In [16], historical arrival time, departure time and temporal feature are used to predict next-day arrival and departure time using Support Vector Machine (SVM) and claimed SMAPE of 2.9% and 3.7% respectively.

In literature, numerous machine learning algorithms are applied for prediction of session duration and energy consumption either on historical or, residential or public charging datasets. Some of them use weather data and vehicle information, nearby events and public holidays. In public charging datasets, user behavior is highly stochastic and uncertain whereas in residential or enterprise charging, user habits are more deterministic, so they are easy to predict. In this work, public charging dataset is considered with weather data, nearby event data and public holiday to consider external factors on user behavior, generalize the model and make them robust. The key contributions of this paper are as follows:

- 1) An approach to predict energy consumption and session duration with reduced MAE. This approach utilizes the predicted session duration as a new feature for the prediction of energy consumption. It also uses local weather data for the prediction. This work highlights the effect of data processing and feature engineering that has improved the results significantly over the existing literature.
- 2) Training, testing and validation of various machine learning algorithms i.e RF, SVM, XGBoost, and ensemble learning models. The performance is compared with the existing work utilizing various models.

II. BACKGROUND

This section gives an overview of machine learning algorithms and their evaluation metrics.

A. Machine Learning

Machine learning is a subset of artificial intelligence that enables computer to learn from past experiences without being explicitly programmed. Machine learning is being used in many applications now days. If any problem can be solved by mathematical modeling then machine learning models are not much useful as mathematical modelling gives error-free and exact solution but when the mathematical modelling of the problem is not easy or not possible in such cases machine learning is very useful where we write algorithms and let algorithm discovers approximate models for complex mappings between input and output.

In supervised learning, the algorithm takes labeled data for training for which the output is well-defined. The algorithm discovers the underlying trends and patterns from the labeled data and maps input-output relationship and selects optimum model. The model can be used for prediction after being trained. It takes unlabeled data and predicts labels as in Fig. 2. In this paper, charging dataset, weather information and nearby events are used as input data. They are also called as features. Session duration and energy consumption are considered as output parameters also called as labels.

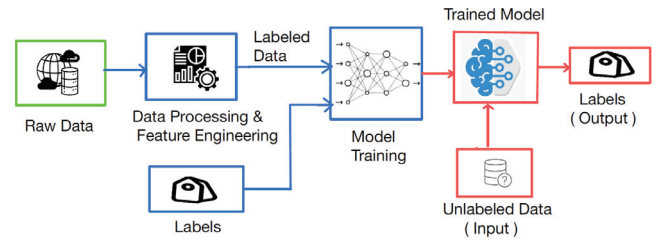


Fig. 2. A typical supervised machine learning framework

In this work, RF, XGBoost, SVM, and ensemble learning models (voting regressor and stacking regressor) are used. RF has decision trees as base models which are strong learners. Therefore, the meta-model in RF reduces the variance and avoids overfitting. XGBoost is an improved version of Gradient Boosting (GB) algorithm which optimizes the hardware as well as software use over gradient boosting algorithm. SVM with kernel trick is used to separate nonlinear data. Ensemble learning methods combine multiple algorithms to obtain better performance than that could be obtained from the individual model. Final predictions are more robust and less prone to error.

B. Evaluation Metrics Regression

Evaluation of regression models can be done using different metrics. Which tells the goodness of the model. The popularly used evaluation metrics with regression are listed below:

1) R^2 regression score: R^2 score will be 1.0 for perfect model and it can be 0 or even negative for worst model. It is calculated as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (1)$$

2) *Mean absolute error:*

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

3) *Symmetric mean absolute error:*

$$SMAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)} * 100\% \quad (3)$$

4) *Root mean absolute error:*

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (4)$$

where y represents the actual value, $\hat{y}$ is the predicted value, $\bar{y}$ is the average of the actual values and n represents the number of test samples in test dataset.

Lower scores of MAE, SMAPE and RMSE generally indicate low error in prediction and the model is considered as a good representative of the dataset. Most of the literature used SMAPE which is more suitable for EV charging prediction because it considers both actual and predicted values in the denominator. In addition to SMAPE, MAE and RMSE are also considered.

III. METHODOLOGY

This section covers problem formulation, exploratory data analysis, model training and validation. A complete flowchart for data preparation and prediction is shown in Fig. 4.

A. EV User Behavior

As stated before, the session duration or departure time for a given connection time and energy consumed by EV during the session are two important parameters of EV user behavior which are considered in this work.

EV user behavior objective can be formulated as follows:

$$EV_{ub} \triangleq (t_{con}, t_{discon}, e_{con}) \quad (5)$$

From (5), the session duration can be calculated as follows:

$$T_{sd} = t_{discon} - t_{con} \quad (6)$$

where t_{con} represents the connection time of the vehicle, t_{discon} is disconnection time of the vehicle, e_{con} is the energy consumed by the vehicle, T_{sd} denotes session duration of the vehicle and EV_{ub} represents user behavior.

B. Dataset Description

The charging dataset used in this work is acquired from a site located at Caltech, Pasadena, USA. In this site currently, the data is collected for 54 Electric Vehicle Supply Equipment (EVSE), which are open to public, staff and students for charging their vehicles. It consists of two main sites one is Caltech site and the other is Jet Propulsion Laboratory (JPL). Unlike Caltech site, JPL site is only open for staff [17]. Therefore, the JPL dataset has not been used in this work.

Weather data is taken from the weather station located at the Caltech USA [19] and from NASA's Modern-Era

Retrospective Analysis for Research and Application, version-2 (MERRA-2) [18], which is collected by the satellite. The satellite data is not that much accurate compared to the data collected by the local station [20]. Therefore, the weather data from local station is considered. But, it has some missing values and important variables such as snowfall, rain etc. So, the missing values and variables are added to the local station data from MERRA-2 weather data, and used in this work. The parameter 'Datetime' of the weather data is in UTC format. So, it is converted to local timezone, Los Angeles, CA which is GMT-7. Nearby events data is taken from Caltech university website calendar [21] to observe the impact of events on user behavior. It is observed that sessions recorded on weekdays are significantly more than the weekends from Fig. 3. So a feature can be created which will assist models to differentiate between weekdays and weekends.

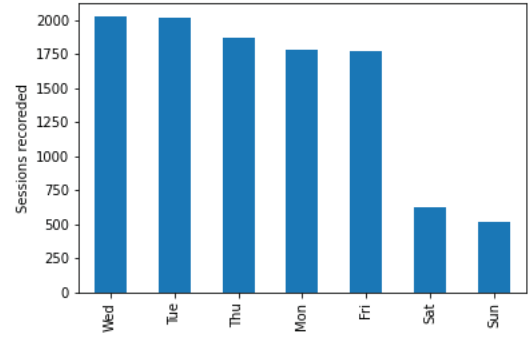


Fig. 3. Number of sessions grouped by weekday

C. Data Cleaning and Pre-processing

Data cleaning and processing play a vital role in generalization of the models. It is important to ensure that the dataset should not contain any missing values as most of the algorithms can't work with missing data. Therefore, missing entries are dropped in this work.

Outliers can negatively impact the performance of the models so identifying and removing those outliers are necessary to learn generalized and robust machine learning models. Outliers are those data points that differ greatly from the other observation in the sample of a population. A most common technique to detect outliers is boxplot of the variables but in case of multivariate models, multivariate outlier detection (anomaly detection) is useful. Isolation forest is one of the most popular multivariate outlier detection algorithms which is a tree-based algorithm. It splits data continuously at random thresholds till all the points get isolated. The outliers are far from other data points so they get separated early. Visualization of the outliers using isolation forest is presented in Fig. 5.

For the charging dataset, only registered records are considered for which user IDs are available. Records from Jan-2019 to Dec-2019 are considered, which captures the seasonal variation throughout the year. Datetime fields are time zone

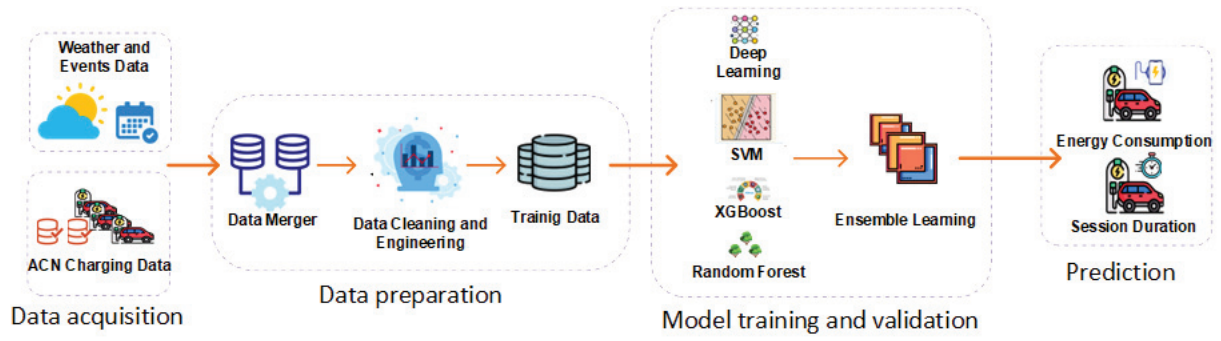


Fig. 4. Complete flowchart for data preparation and prediction of EV users' behavior

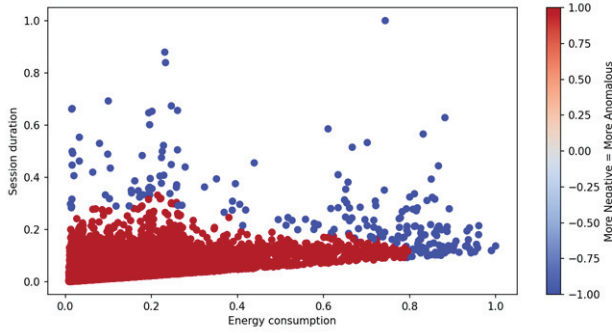


Fig. 5. Multivariate outlier detection

aware with local time, so they are converted to naive datetime objects. Datetime fields of weather data from MERRA-2 and Caltech local station are recorded with UTC (GMT) time so they are converted to be the same as charging dataset which is "America/Los_Angeles" (GMT-7). The last 7 hours' and the next 10 hours' average values of the weather data have been calculated to determine the effect on the charging decisions of EV users. For events data, the total number of events after the connection time of the vehicle to the end of the day is calculated. This allows models to learn how the total number of events affects the charging behavior of the users on that day.

Finally, all the data is merged with the charging dataset by obtaining the data at the nearest hour of connection time from the records.

D. Feature Engineering and Selection

Feature engineering is a process of developing new features by converting raw data into the desired features using statistical or machine learning approach. It consists of feature creation, feature transformation and feature extraction using human knowledge. This is required to make training easier because models are not capable to learn directly from raw data.

The connection time of the records is a datetime object and machine learning model cannot work directly with datetime.

So some new features, weekdays, weekends, holidays according to US federal holidays calendar 2019, day and month from the date information and hour by adding minutes divided by 60 to hour entry from time information are extracted.

As the hour, day and month are cyclic in nature so sine and cosine transformations are used to better represent them. Other way is to use one hot encoding but It will create too many features so it is not used.

$$f_x = \sin(2\pi f / \max(f)) \quad (7)$$

$$f_y = \cos(2\pi f / \max(f)) \quad (8)$$

where f denotes cyclic feature and f_x and f_y represent sine and cosine components of the feature respectively.

The Feature 'userId' is not directly useful. So it is used to create some new features by calculating average connection time, disconnection time, average session duration and average energy consumed for each userId. Some of the algorithms like K-NN, SVM and NN are sensitive to feature value instead of the trends and patterns of it and gives importance to the feature which has a higher value irrespective of their measuring unit. So, one of the most important transformation, feature scaling is done to deal with it. Other advantage is that gradient descent converges faster with scaled features. Most common technique for feature scaling is min-max scale and standardization. Min-max scalar transforms feature between [0,1] whereas the standardization scales features to mean zero and unit variance features. The min-max scalar is used in this work.

E. Model Selection and Evaluation

One of the challenges of applied machine learning is to select the best model among a range of different models. The best model is a good representative of both training and test data. It should perform well on train as well as test dataset. If training error is less but testing error more, than the model is over-fitted. It is not well generalized and it will perform poorly on test dataset. If the model is giving more error even for training dataset, then the model is under fitted in that case, model complexity needs to be increased. Problem of under-fit and over-fit is rectified by trading off between variance and bias of the model by hyper-parameter

tuning. Calendar year-2019 has been selected for the charging dataset ensuring seasonal variations. The dataset is split into 80:20 ratio for training and testing respectively. Grid-search or random-search with K-fold cross-validation is used to avoid sampling bias and, to find the best hyper-parameters. The entire data are divided into K folds and the model is trained for K-1 folds. The process is repeated for all folds and obtained model is evaluated using performance metrics to select optimal parameters based on training and validation employing the most popular and successful algorithms like RF, XGBoost, SVM, and from ensemble learning namely voting regressor and stacking regressor. Scikit-learn from python library is used to train and test the models.

IV. RESULTS AND COMPARISON

Random forest algorithm is trained first as it gives relative feature importance. Then those features have been selected which contribute to the prediction significantly. This is also known as the second round of feature selection. RF, SVM,

TABLE I
TEST SCORE FOR SESSION DURATION

Model	R^2 Score	MAE (mins)	RMSE (mins)	SMAPE (%)
RF	0.70	68.1	100.5	12.67
XGBoost	0.71	67.5	98.0	12.8
SVM	0.67	71.6	105.0	14
Deep ANN	0.67	70.5	105.0	13.4
Voting Ensemble	0.71	66.5	97.9	12.66
Stacking Ensemble	0.70	67.5	99.6	12.6

XGBoost and ANN are trained first for the prediction of session duration, and then predicted session duration is used as a new feature to train the model for prediction of energy consumption. Three best-performing algorithms RF, SVM and XGBoost are selected to form voting ensemble and stacking ensemble models. In stacking ensemble, meta-model used is XGBoost as it performs better than other models. The obtained test scores are summarized in Table I for session

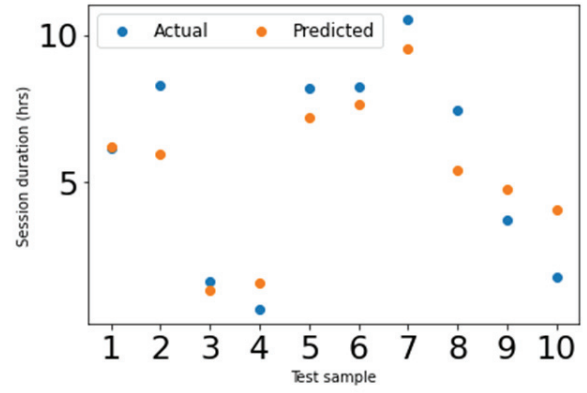


Fig. 6. Prediction for session duration

duration and Table II for energy consumption. The best result is obtained from voting ensemble algorithm as highlighted in their respective tables. The results obtained for ten samples can also be visualized in Fig. 6 for session duration and Fig. 7 for energy consumption.

TABLE II
TEST SCORE FOR ENERGY CONSUMPTION

Model	R^2 Score	MAE (kWh)	RMSE (kWh)	SMAPE (%)
RF	0.74	1.47	2.23	11.18
XGBoost	0.74	1.50	2.20	11.4
SVM	0.67	1.75	2.50	13.35
Deep ANN	0.70	1.58	2.38	11.79
Voting Ensemble	0.75	1.45	2.19	11.0
Stacking Ensemble	0.74	1.48	2.21	11.2

A comparison of this work with the literature is summarized in Table III. Most of the literature uses SMAPE and MAE metrics to evaluate their models. An important observation is that the performance scores are better for residential and enterprise charging, because the users' behavior is routine behavior, which is easy to predict. Whereas, in the case of

TABLE III
COMPARISON WITH RELATED WORKS

Source	Model Used	Dataset Used	Energy Consumption	Session Duration
[10]	UCLA Campus (historical & residential charging data from UK)	Ensemble learning of SVM, RF, DKDE	SMAPE: 7.5%	SMAPE: 10.4 %
[11]	German historical charging data (vehicle and location info.)	XGBoost	-	MAE: 82 mins
[12]	ACN (historical charging data)	SVM, RF, XGBoost, ANN, Ensemble learning	MAE: 3.3 kWh SMAPE: 11.6 %	MAE: 66.5 mins SMAPE: 9.92%
[13]	Nebraska (historical public charging data)	XGBoost, RF, SVM	MAE: 4.6 kWh	-
[14]	UCLA (historical energy data)	K-NN (K=1)	R^2 score: 0.52 SMAPE: 15.3%	-
[15]	ACN (historical charging data)	GMMs	SMAPE: 15.9%	SMAPE: 14.4%
[23]	UCLA (historical energy data)	PSF	SMAPE: 14.1%	-
[24]	Historical charging data (UK and US dataset)	hybrid KDE	MAE: 1.68 kWh	MAE: 45 mins
This work	ACN (historical charging data) (weather and campus events)	XGBoost, RF, SVM ANN, Ensemble learning	MAE: 1.45 kWh SMAPE: 11.0	MAE: 66.5 mins

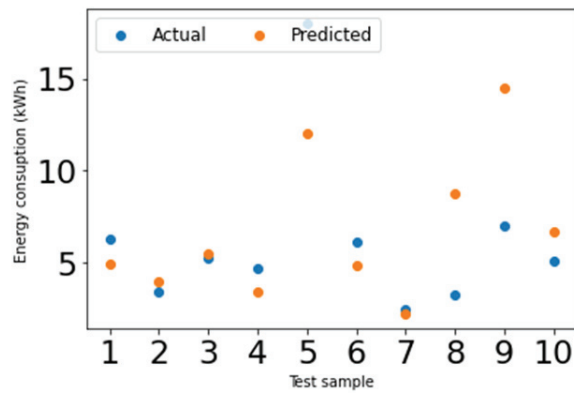


Fig. 7. Prediction for session duration

public charging dataset, it is more difficult to predict the behavior because of highly uncertain behavior of the users. This work is for the public charging dataset, so it is compared with the literature that uses public charging dataset. Different from the past work, the predicted session duration is used for prediction of the energy consumption. Instead of satellite weather data, the weather data from the local station is used for prediction. The performance metrics score for the energy consumption, i.e. MAE and SMAPE is significantly improved over the existing literature for the same charging dataset [12], [15], whereas, for the session duration, MAE is as good as reported in literature [12] with an improved data processing pipeline.

V. CONCLUSION

In this work, two most important EV user behavior parameters, i.e. energy consumption and session duration are predicted for the public charging dataset. It is observed that among the various algorithms that are shown, the voting ensemble gives the best result. Unlike the work in existing literature, this work uses the predicted session duration as a new feature for the energy consumption prediction. Also, it uses the weather data from local station and campus events for the prediction. It is shown that the result obtained for the energy consumption is significantly improved over the past efforts, while the session duration remains as good as that reported in the literature [12]. The results can be improved further by including vehicle information (such as battery charge, capacity, etc.), traffic data, more nearby events data, and deep learning models that take time-series into consideration. The predicted user behavior can be used in future for the smart charge scheduling of EVs.

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